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Overview of Drone Technology

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1.1 Introduction

Unmanned aerial vehicles (UAVs), commonly known as drones, have emerged as a pivotal innovation in modern aerospace technology, transforming the way aerial operations are conducted across military, commercial, and recreational domains. Initially developed for military reconnaissance and surveillance during the early 20th century, UAVs have since evolved into highly sophisticated systems equipped with advanced sensors, navigation units, and autonomous control capabilities. Their rapid technological growth has enabled applications far beyond defense, reaching industries such as agriculture, logistics, construction, environmental monitoring, and disaster management.

In the commercial sector, drones are revolutionizing data acquisition, aerial mapping, and delivery systems through cost-effective and efficient aerial solutions. The recreational use of drones has also surged, offering enthusiasts access to aerial photography, racing, and hobbyist exploration, supported by improved flight stability and user-friendly interfaces.

The core of drone technology lies in the seamless integration of aerodynamics, propulsion, communication, navigation, and artificial intelligence. Continuous advancements in battery technology, miniaturized sensors, GPS precision, and autonomous flight algorithms are driving the